PPO for BTC/USDT

Methodology, optimisation and evaluation

Project_PPO_All_In_One_Fixed.ipynb

Evaluation: 17 August–17 September 2026

End exclusive · 744 candles · 743 transitions

Research question and baseline

Can policy selection improve BTC exposure decisions?

Hourly BTC/USDT observations; nine discrete exposure choices.

Baseline: three PPO networks combined by **mean probability**.

Optimisation: features, update settings, turnover penalties and aggregation.

Assessment: chronological validation, followed by a fixed evaluation window.

“Selected” means best eligible PPO validation score. It does not mean highest evaluation return.

Source: fixed notebook and captured experiment; all dates UTC.

Why PPO

A categorical actor directly represents the nine action probabilities.

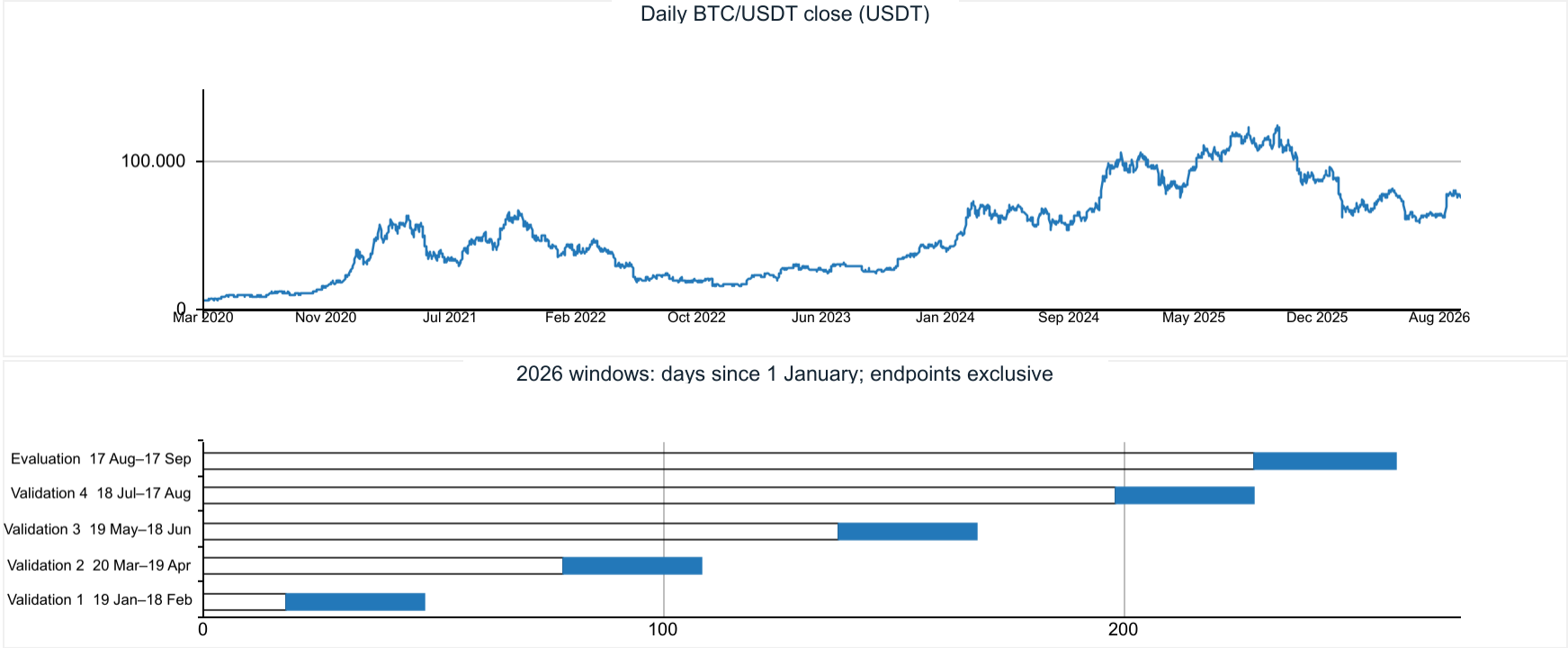
A critic estimates state value to support advantage-based learning.

Clipping, gradient limits and a KL check control policy updates.

Fresh rollouts are reused for a few epochs, then discarded.

A reasonable research choice; no evidence here that PPO is universally best for trading.

Data and chronological windows



Daily closing prices are plotted from the verified hourly dataset. Shaded windows show four 30-day validations and the 31-day evaluation. Date endpoints are exclusive; all timestamps are UTC.

Source: fixed notebook and captured experiment; all dates UTC.

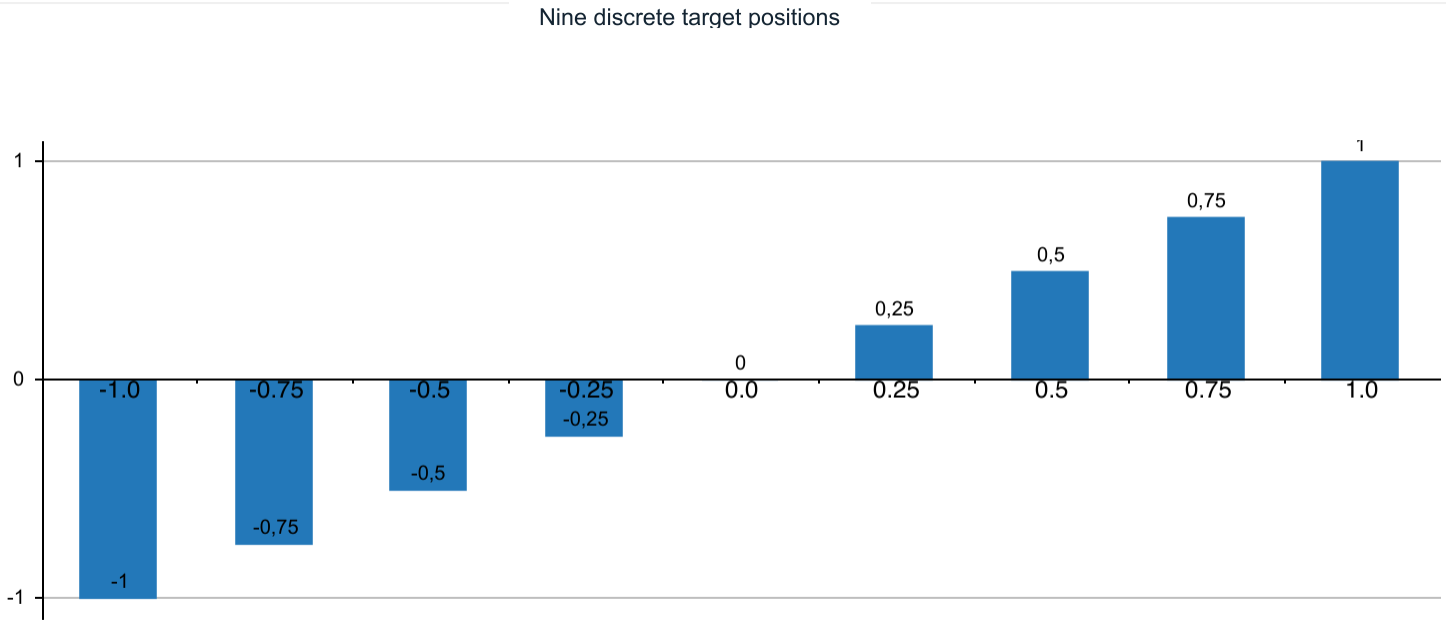
State representation

Input	Baseline	Momentum alternatives
Static market features	5	13
Main information	Hourly return, price ratios, relative volume	Multi-horizon returns, volatility, trends, candle shape, relative volume
Dynamic features	Previous target position and current realised exposure	
Total input dimensions	7	15

Fit the scaler on each training partition; apply it unchanged to validation and evaluation.

Source: fixed notebook and captured experiment; all dates UTC.

Nine discrete target positions



Position values are target exposures, not buy/sell quantities or a continuous action space. Repeating a target does not force rebalancing in the installed simulator; realised exposure can drift.

Source: fixed notebook and captured experiment; all dates UTC.

Network and training architecture

Observation → separate actor and critic networks

Actor	7 or 15 inputs → 64 tanh → 64 tanh → 9 logits
Critic	7 or 15 inputs → 64 tanh → 64 tanh → 1 value
Rollout batch	4 environments × 256 steps = 1,024 transitions
Update	GAE; normalised advantages; 6 epochs; minibatches of 256
Evaluation	Frozen weights; deterministic action selection; no learning

Source: fixed notebook and captured experiment; all dates UTC.

Reward and simulation assumptions

$$r_{train} = 100 \times [\log(V_{t+1}/V_t) - \kappa\,|\Delta target|]$$

- Initial portfolio: **1,000 USDT in cash**.
- Trading-fee parameter: **0.01%**; hourly borrowing parameter: **0.0003%**.
- Turnover penalty κ shapes learning; it is not an extra portfolio debit.
- Reported return comes from simulated portfolio value, not shaped reward.

Costs are incomplete: no spread or slippage; borrowing accumulation has a verified defect.

Source: fixed notebook and captured experiment; all dates UTC.

Optimisation hypotheses

Candidate	Change	Intended effect
Momentum	13 features + entropy + annealing	More market context
Cautious	Lower rate; tighter clip	Smaller policy updates
Short horizon	$\gamma = 0.95$; $\lambda = 0.90$	Less distant reward dependence
Turnover / low turnover	$\kappa = 0.0003$ / 0.0001	Fewer target changes
Exploration	Entropy coefficient 0.03	More action diversity

These are configuration bundles. Their individual components were not isolated in ablations.

Source: fixed notebook and captured experiment; all dates UTC.

Experimental design

Stage	Fits	Steps per fit	Transitions
Screening: 7 configs × 4 folds × 1 seed	28	16,384	458,752
Confirmation: 3 configs × 4 folds × 3 seeds	36	49,152	1,769,472
Final refit: 1 config × 3 seeds	3	49,152	147,456
Total	**67**		**2,375,680**

Then compare 2 aggregation rules × 3 candidates × 4 folds.

Seeds: 23, 42, 71. Selection is frozen before final evaluation.

Source: fixed notebook and captured experiment; all dates UTC.

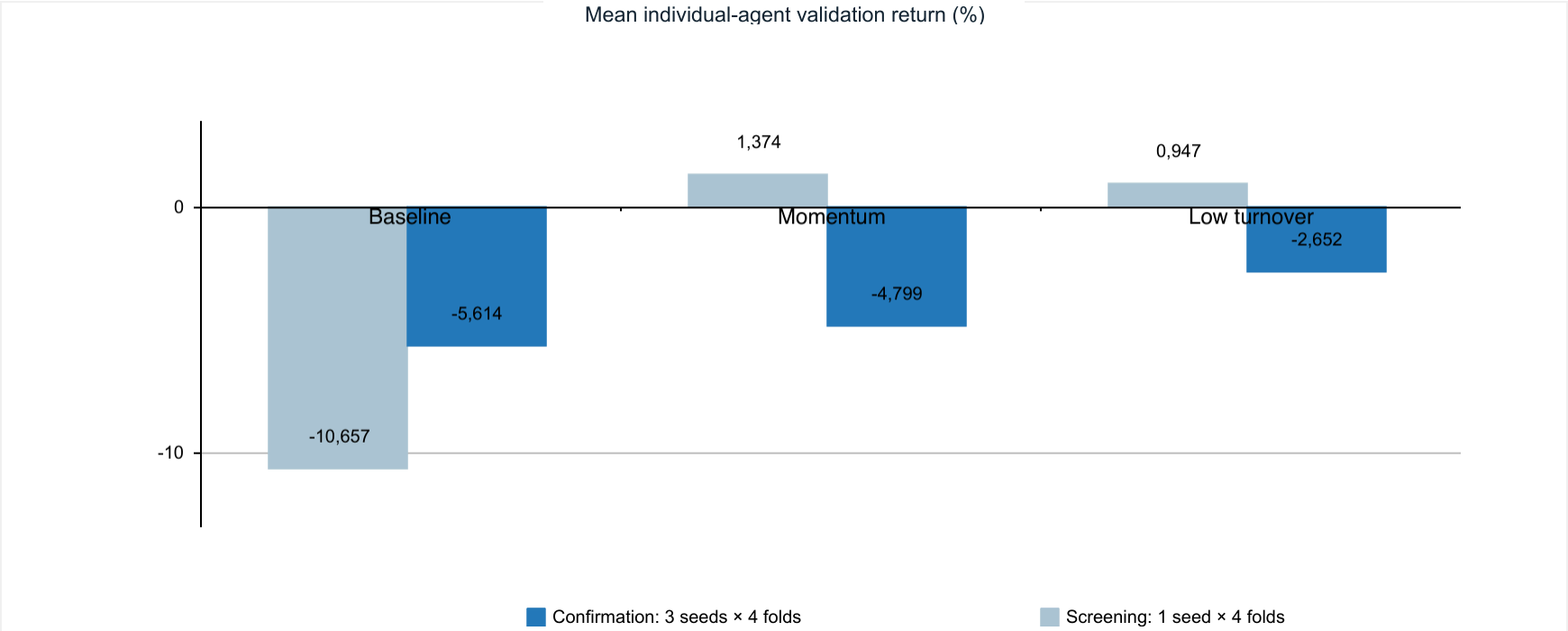
Initial screening results



Each point is one validation fold. Bars are arithmetic means of four fold returns, seed 23, 16,384 training transitions per fit. These are validation results, not final evaluation returns.

Source: fixed notebook and captured experiment; all dates UTC.

Confirmation reduced confidence in the initial gains



Confirmation refits each candidate from scratch using 49,152 transitions and seeds 23, 42 and 71. Both seed coverage and training budget changed, so the shift is not a controlled estimate of either effect alone.

Source: fixed notebook and captured experiment; all dates UTC.

Mean probability and unanimous vote

Baseline: mean probability

Average each action's probability across the three networks.

Choose the action with the largest average.

$$\operatorname{argmax}_a \operatorname{mean}_i \pi_i(a \mid s)$$

Selected: unanimous vote

Find each network's most likely action.

Execute only if all three agree exactly; otherwise choose cash.

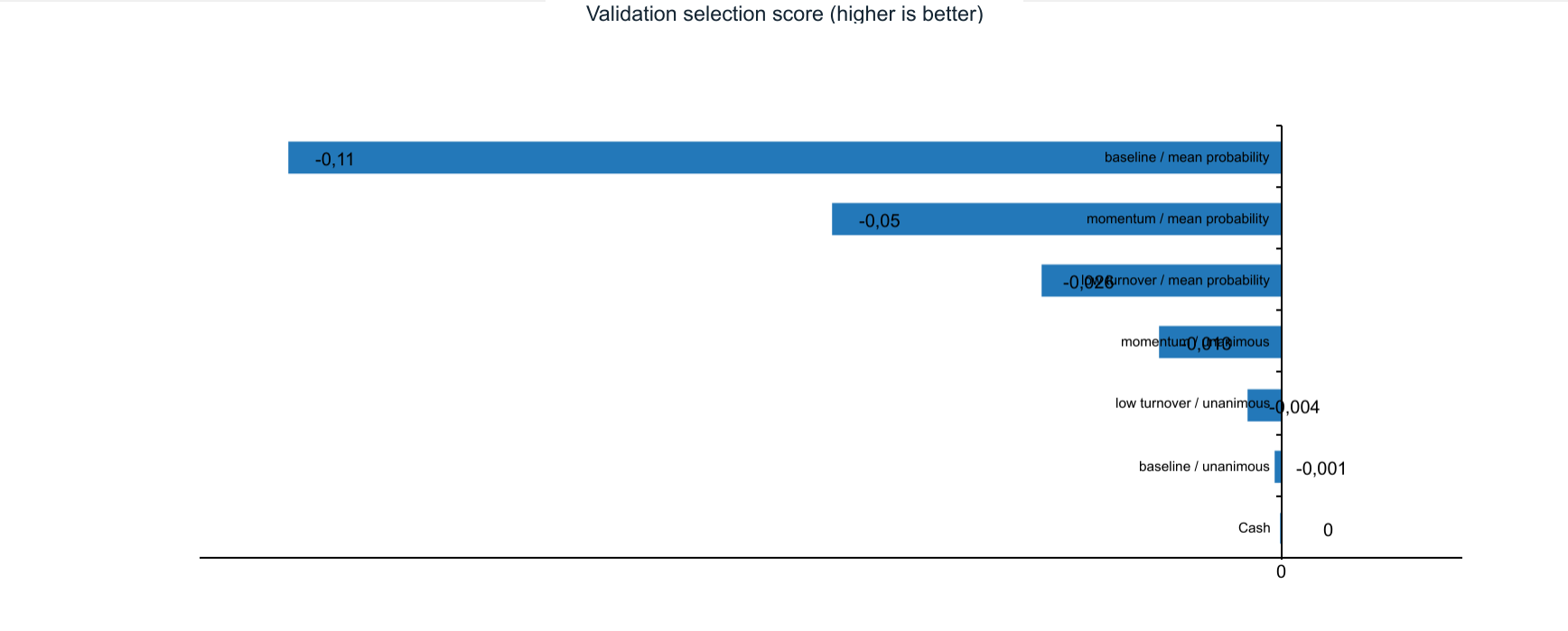
All votes equal → common action

Disagreement → cash

Both final policies use the same three baseline network weights.

Source: fixed notebook and captured experiment; all dates UTC.

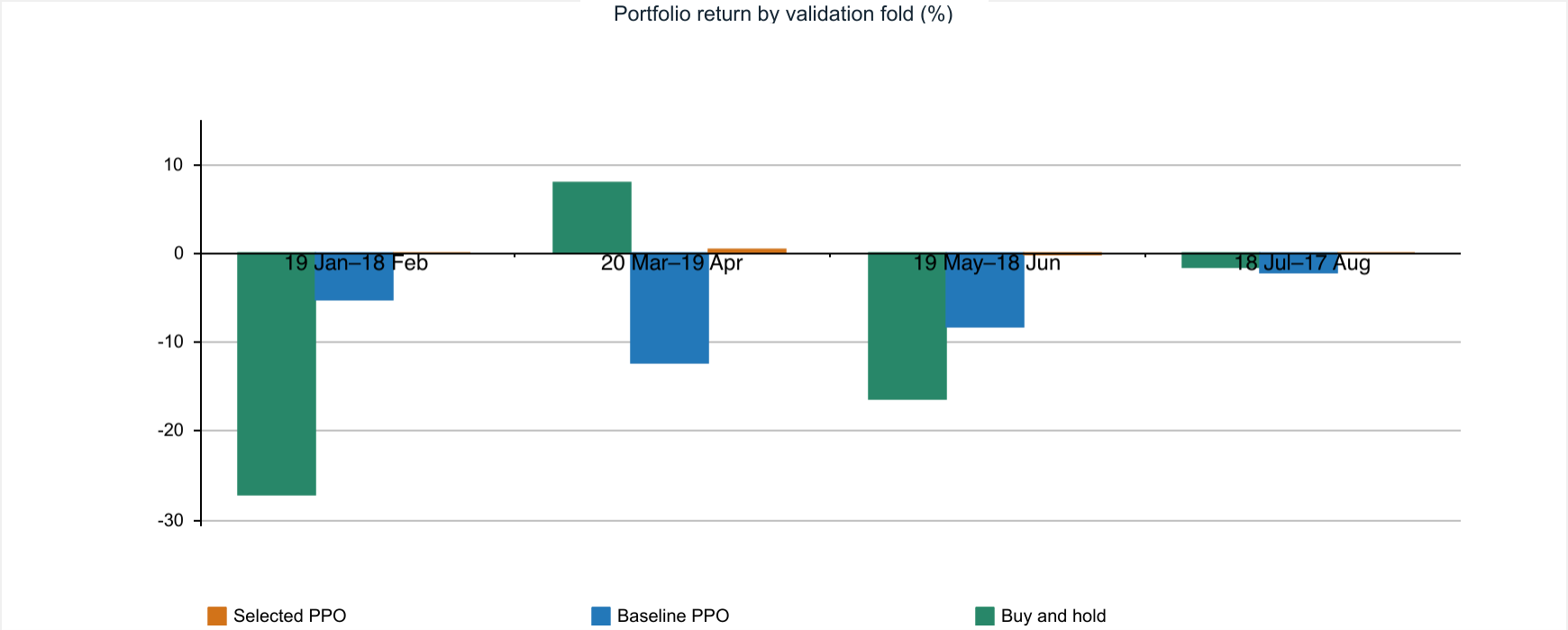
The validation score selected abstention



Score = mean(log return) − 0.25 × mean(drawdown magnitude) − 0.10 × sample standard deviation(log return). Scores use four ensemble backtests, one per fold. Cash is shown for context but was excluded from selectable PPO candidates.

Source: fixed notebook and captured experiment; all dates UTC.

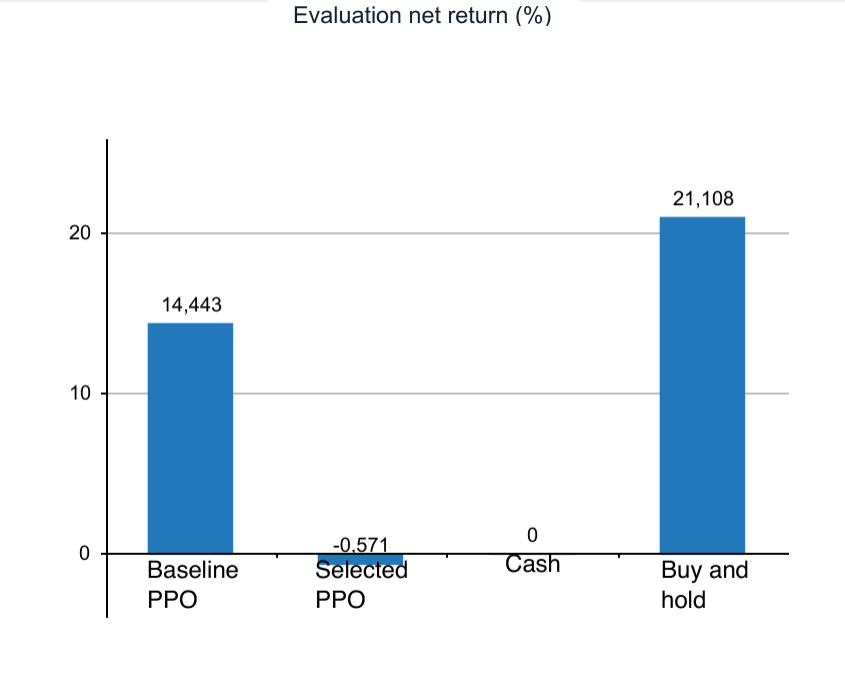
Validation and evaluation had different market patterns

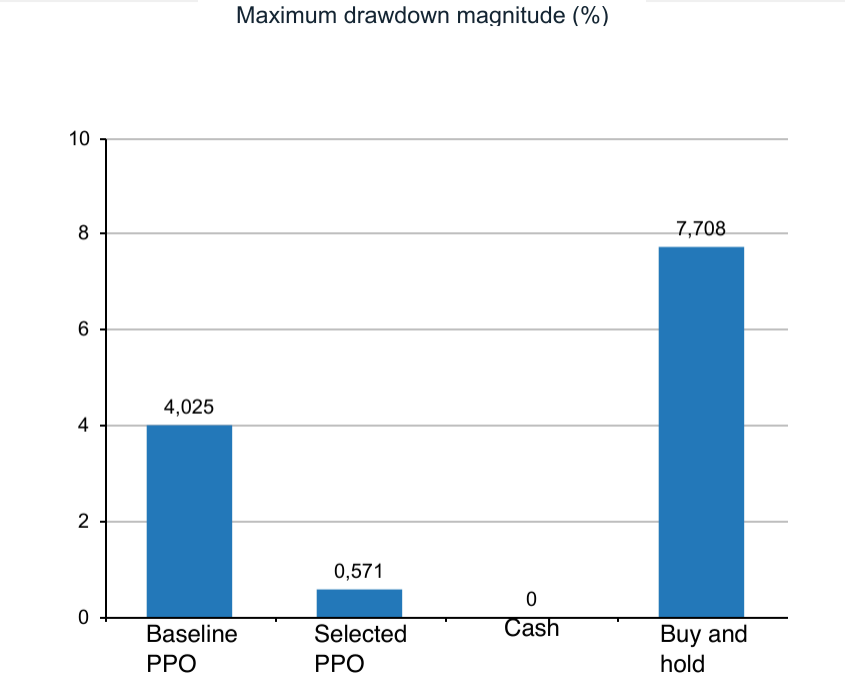


All dates are in 2026 UTC, end exclusive. Each fold has 720 observations and 719 transitions. The selected policy's fold returns were 0.00%, +0.48%, −0.24% and −0.09%.

Source: fixed notebook and captured experiment; all dates UTC.

Baseline beat selected PPO during evaluation

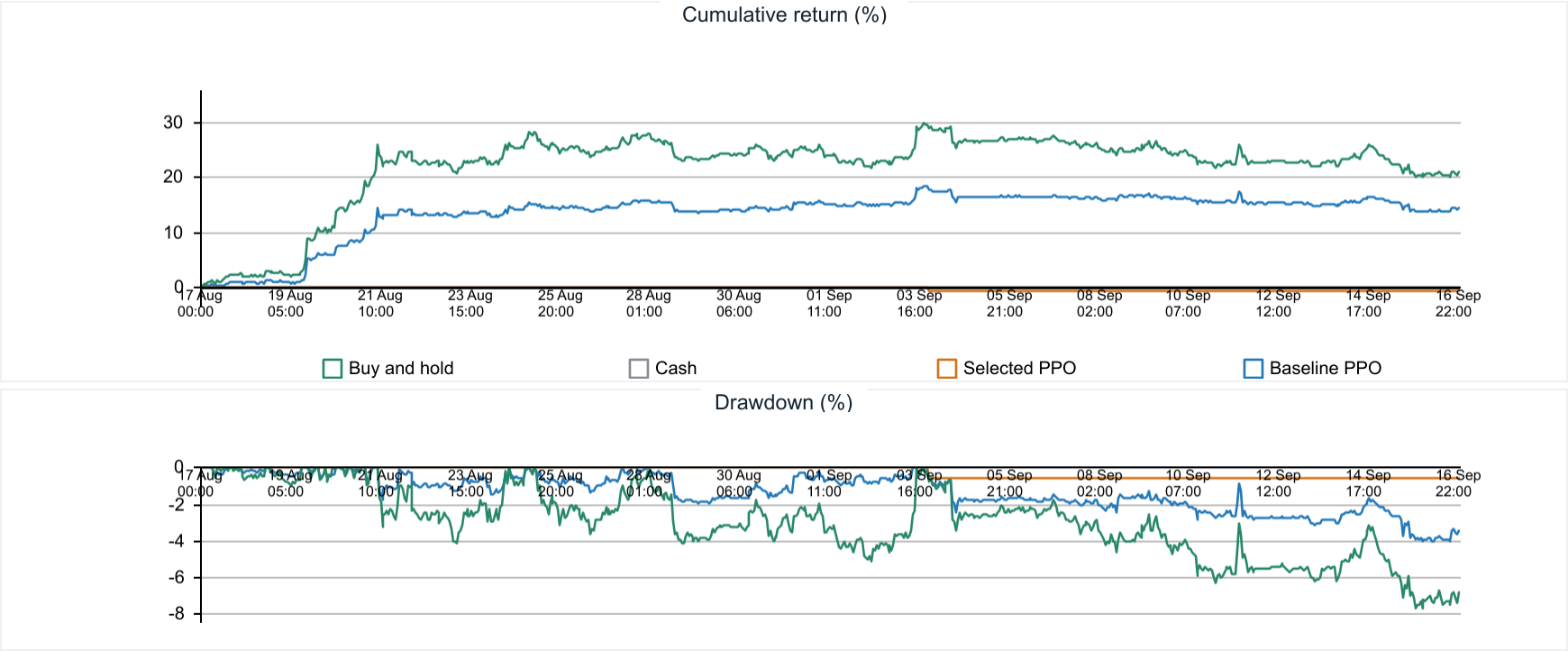




Evaluation candle labels: 17 Aug 2026 00:00 through 16 Sep 2026 23:00 UTC. Returns use first-to-last included close, with simulator fees. Borrowing accounting has a verified limitation; slippage and spread are omitted.

Source: fixed notebook and captured experiment; all dates UTC.

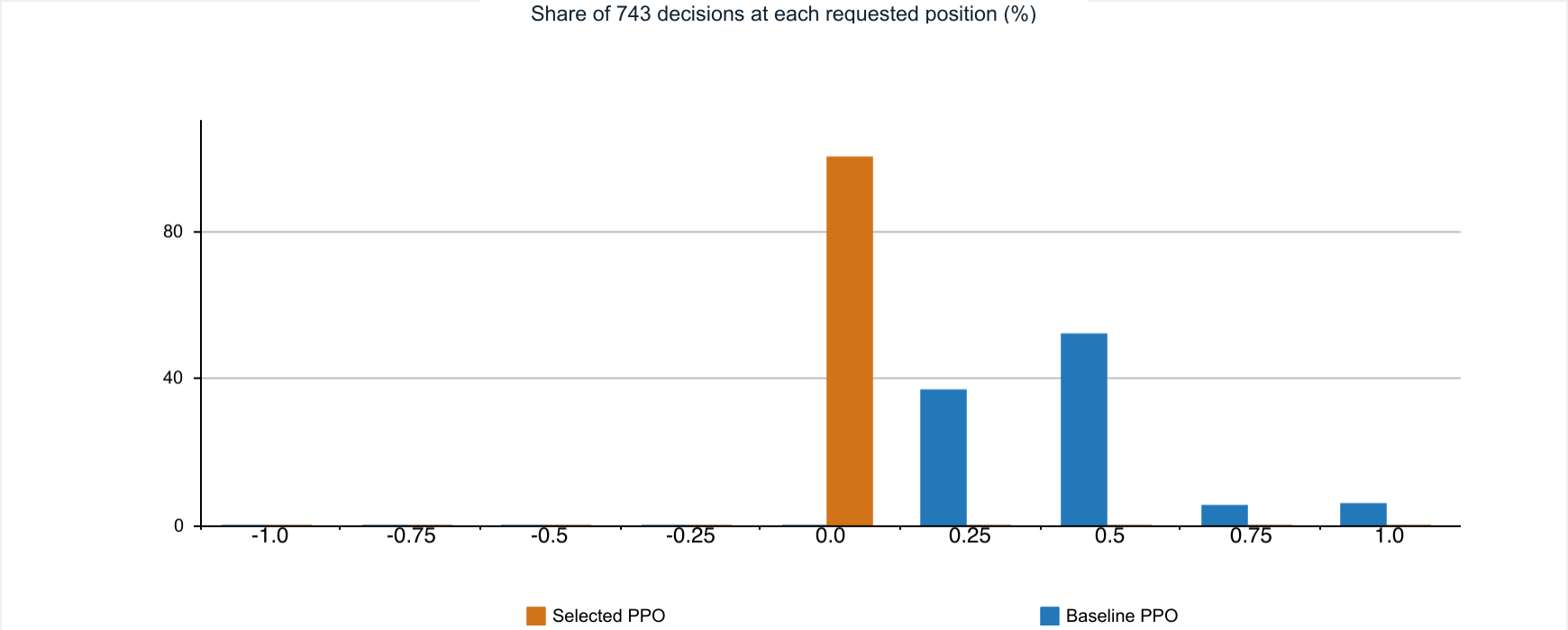
Portfolio paths and drawdowns



All paths start at 1,000 USDT in cash. Selected PPO ends at 994.29; baseline at 1,144.43. Curves are recomputed from the live trained agents and exactly match the stored experiment traces.

Source: fixed notebook and captured experiment; all dates UTC.

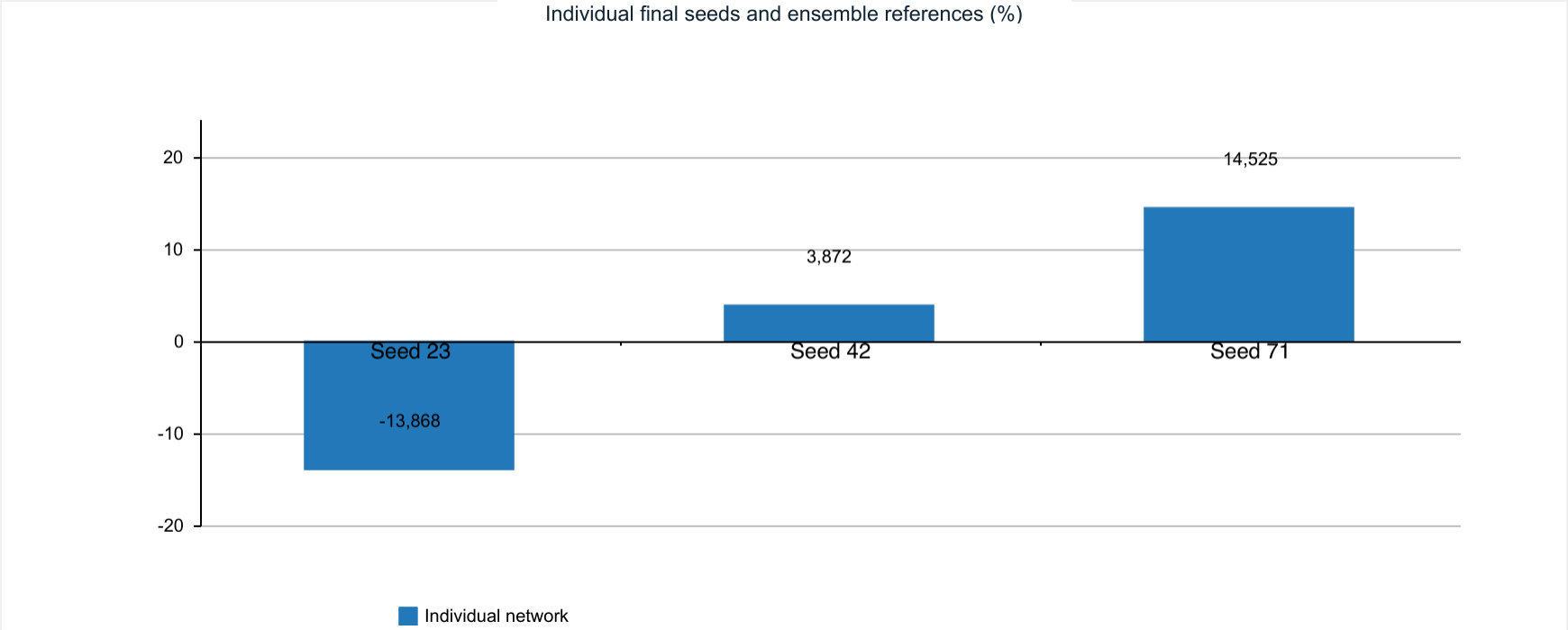
Low drawdown came from almost no market exposure



Direct replay confirms all 742 cash decisions were caused by disagreement between the three models. They unanimously chose +1 only once, producing one holding interval and two position changes (entry and exit). These are decision frequencies, not capital weights.

Source: fixed notebook and captured experiment; all dates UTC.

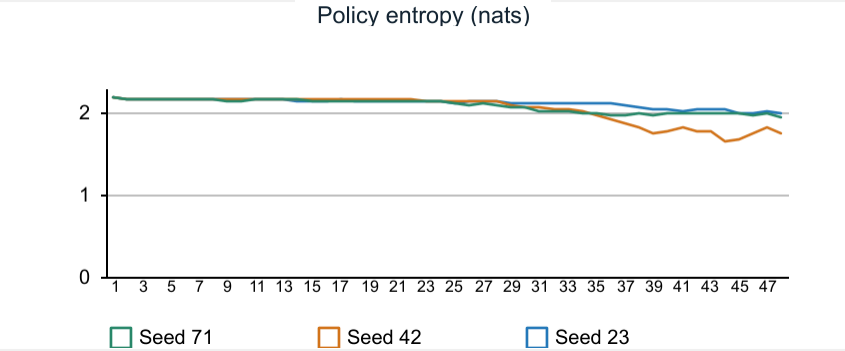
Seed sensitivity remains material

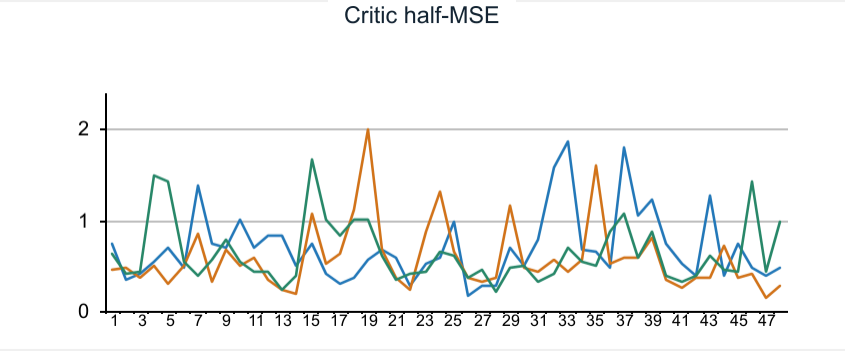


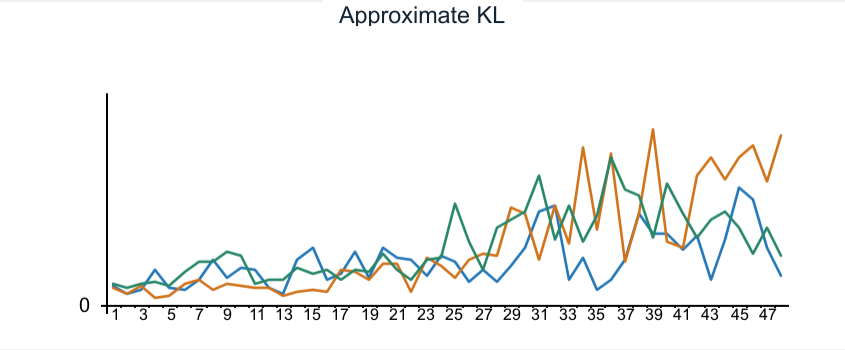
These are the same three fitted networks used by both ensembles. The six notebook rows repeat these three members under two policy labels; they are not six independent models. An ensemble combines decisions, not realised returns.

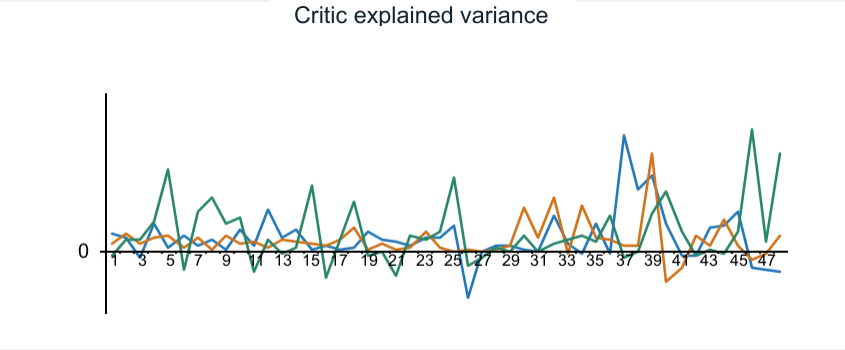
Source: fixed notebook and captured experiment; all dates UTC.

Learning diagnostics do not prove convergence









Final baseline networks: 48 rollout batches × 1,024 transitions = 49,152 per seed; 1,152 minibatch optimiser updates per seed. No KL early stop occurred. Entropy declines, but noisy critic explained variance does not establish predictive skill or convergence.

Source: fixed notebook and captured experiment; all dates UTC.

Limitations that affect the conclusion

Financing defect: interest is overwritten rather than accumulated on a held short.

Optimistic execution: same-close trading; no spread, slippage or latency.

Selection risk: only four recent folds; the evaluation month was reused.

Weak independence: ten repeated episodes replay the same deterministic path.

Attribution: bundled changes prevent isolated causal conclusions.

Source: fixed notebook and captured experiment; all dates UTC.

Next experiments: improve credibility before complexity

1. Correct financing and specify rebalancing and liquidation rules.
2. Test feasible execution with measured cost scenarios.
3. Allow cash and buy-and-hold to compete in policy selection.
4. Freeze choices before a new untouched evaluation period.
5. Run matched ablations and test less restrictive ensemble rules.

These are proposed experiments. No improvement from them has been demonstrated yet.

Source: fixed notebook and captured experiment; all dates UTC.

Conclusion

The search reduced risk mainly by choosing cash.

Baseline +14.44% **Selected −0.57%** **Buy-and-hold +21.11%**

A complete PPO pipeline ran 67 fits with chronological selection.

The selected aggregation improved validation score and reduced activity.

It did not improve evaluation return over baseline or cash.

Robust trading advantage remains unproven.

Source: fixed notebook and captured experiment; all dates UTC.

References and evidence

Primary evidence: the fixed notebook, its live frozen agents, trial tables and hourly portfolio traces.

Schulman et al. *Proximal Policy Optimization Algorithms* (2017).
https://arxiv.org/abs/1707.06347

Schulman et al. *Generalized Advantage Estimation* (2015; revised 2018).
https://arxiv.org/abs/1506.02438

Binance public-data documentation.
https://github.com/binance/binance-public-data

Gym Trading Environment documentation and installed version 0.3.5.
https://gym-trading-env.readthedocs.io/en/latest/environment_desc.html